

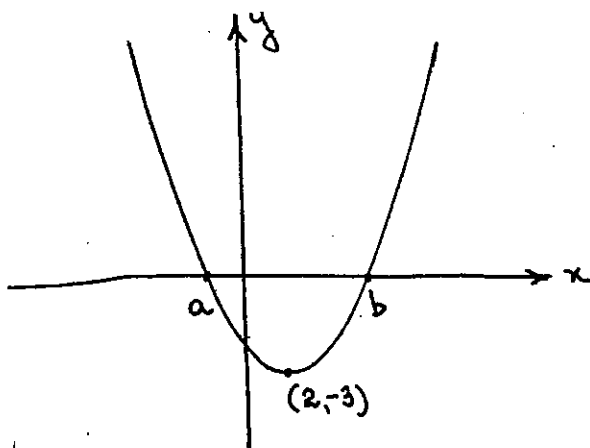
Time allowed: 70 minutes.

All questions **may** be attempted.

START EACH QUESTION ON A NEW PAGE

Question 1 (10 marks)

a. The graph of $y = f(x)$ is sketched in the diagram shown.



Copy this graph and draw neat sketches, on separate diagrams, showing all the main features, of

(i) $y = |f(x)|$ 1

(ii) $y = f(-x)$ 1

(iii) $y = \frac{1}{f(x)}$ 2

(iv) $y = [f(x)]^2$ 2

(v) $y = \sqrt{f(x)}$ 2

b. Draw a neat sketch, showing the main features, of the graph of $y = \cos(\sin^{-1} x)$. 2

Question 2 (15 marks)

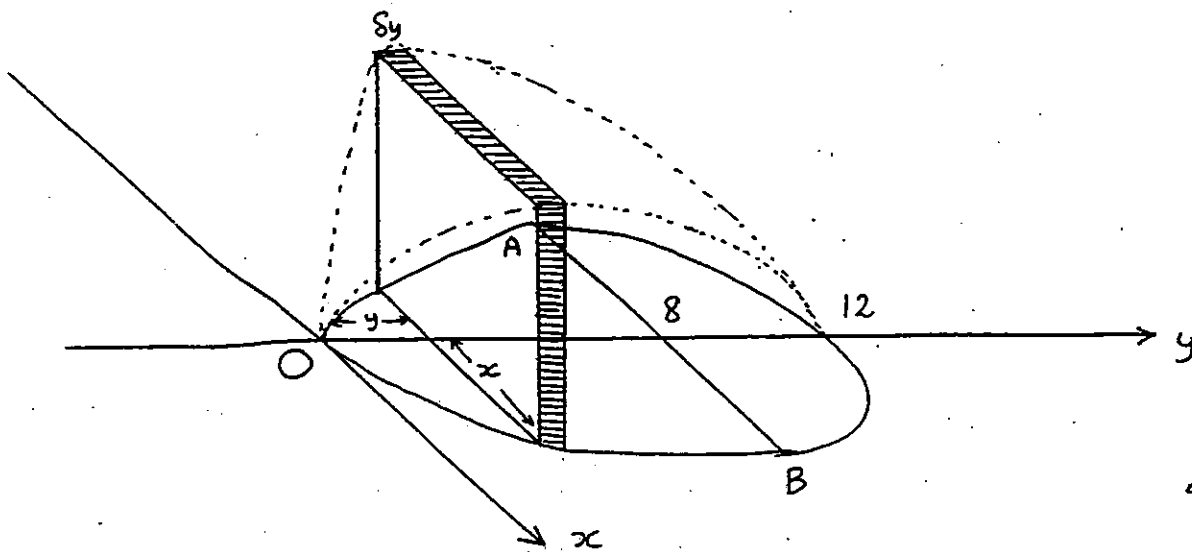
a. The region R in the first quadrant bounded by the curve $y = x(4 - x^2)$ and the x -axis is rotated about the line $x = 2$ to form a solid of revolution. Use the method of "cylindrical shells" to find the volume of the solid.

4

b. The region bounded by $y = 4x - x^2$ and $0 \leq x \leq 4$ is rotated about the y -axis. The solid so formed is sliced by planes perpendicular to the y -axis. Express the areas of the cross-sections so formed as a function of y and hence find the volume of the solid.

5

c.



The diagram shows a solid with its base in the x - y plane. Every cross section perpendicular to the y -axis is a square. One part of the base is the segment OAB of the parabola $x^2 = 2y$ cut off by the line $y = 8$. The other part of the base is a semicircle with diameter AB .

Consider a slice of the solid of width δy and perpendicular to the y -axis as shown.

(i) Find an expression for the volume δV of S in terms of x and δy .

1

(ii) Find the volume of the solid.

5

Question 3 (18 marks)

a. Using the substitution $u = x + 2$, or otherwise, find $\int \frac{dx}{\sqrt{x^2 + 4x + 3}}$. 2

b. Find $\int \frac{dx}{e^x + e^{-x}}$ using the substitution $u = e^x$. 2

c. Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$ using $t = \tan \frac{x}{2}$ 3

d. Using partial fractions, find $\int \frac{6x + 4}{(x + 2)(4 - 3x^2)} dx$ 4

e. Find $\int \tan^4 x \, dx$ 3

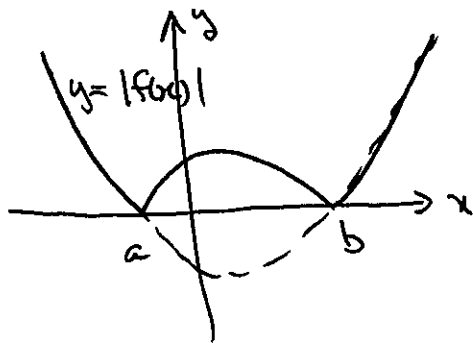
f. (i) If $I_n = \int_1^e x(\ln x)^n dx$ for n a positive integer show that

$$I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1} \quad 3$$

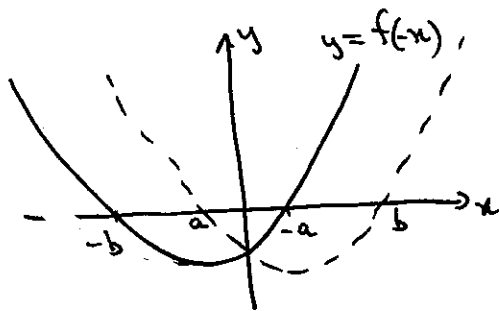
(ii) Hence evaluate $\int_1^e x(\ln x)^3 dx$ 1

END OF EXAMINATION

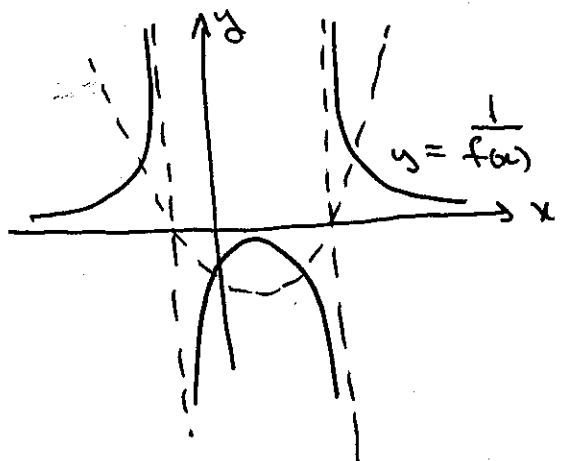
Q. 1 a (i)



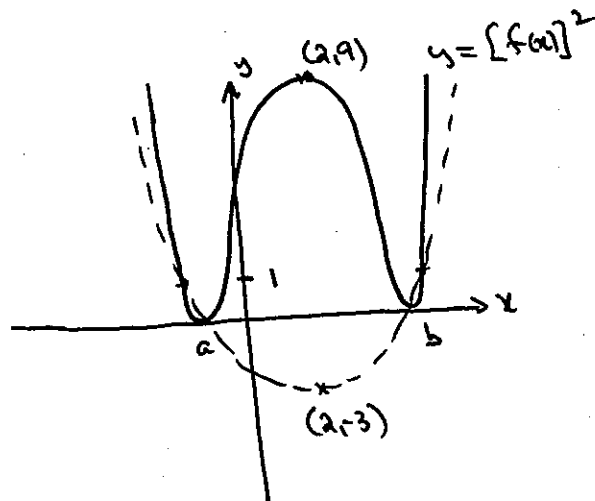
(ii)



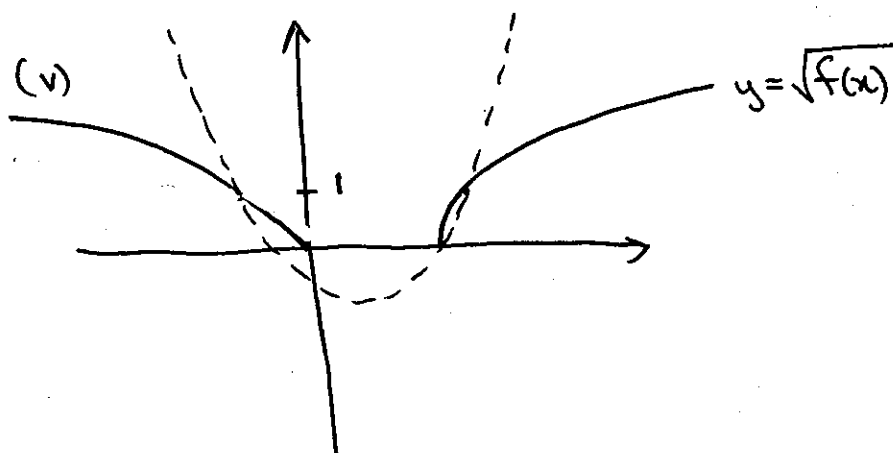
(iii)



(iv)



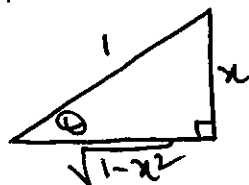
(v)



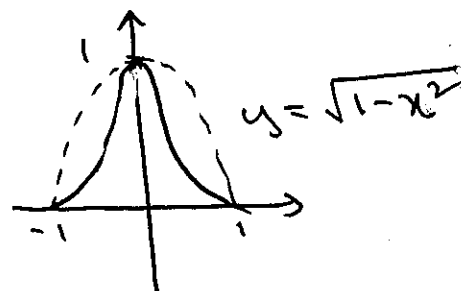
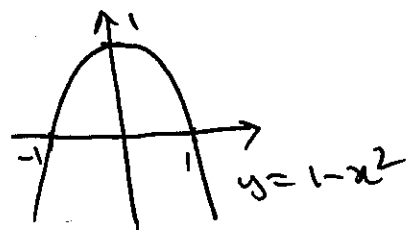
b) $y = \cos(\sin^{-1} x), -1 \leq x \leq 1$

Let $\theta = \sin^{-1} x, -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

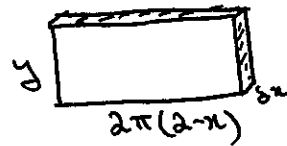
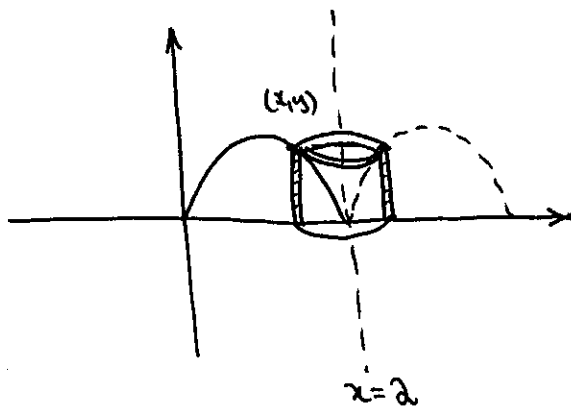
then $\sin \theta = x$



$$y = \cos \theta = \sqrt{1-x^2}$$



Q.2 a.



$$\delta V = 2\pi(2-x)y \delta x$$

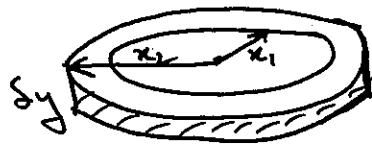
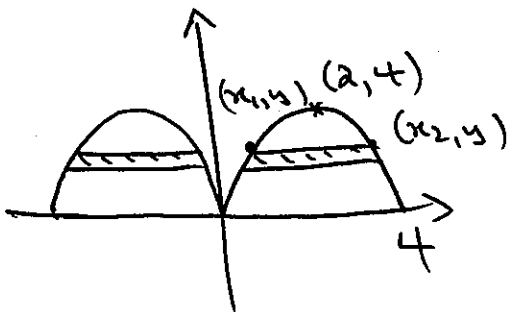
$$V = \int_0^2 2\pi(2-x)(4x-x^2) \cdot dx$$

$$= 2\pi \int_0^2 (8x - 4x^2 - 2x^3 + x^4) \cdot dx$$

$$= 2\pi \left[4x^2 - \frac{4x^3}{3} - \frac{x^4}{2} + \frac{x^5}{5} \right]_0^2$$

$$= \frac{112\pi}{15}$$

b.



$$\delta V = \pi(x_2^2 - x_1^2) \delta y$$

$$\therefore V = \int_0^4 \pi(x_2 + x_1)(x_2 - x_1) \cdot dy$$

$$= \int_0^4 \pi 4 \sqrt{16 - 4y} \cdot dy$$

$$= -\pi \left[\frac{2}{3} (16 - 4y)^{3/2} \right]_0^4$$

$$= \frac{128\pi}{3}$$

$$y = 4x - x^2$$

$$\text{i.e. } x^2 - 4x + y = 0$$

$$\therefore x_2 + x_1 = 4$$

$$(x_2 - x_1)^2 = (x_2 + x_1)^2 - 4x_2x_1$$

$$x_2 - x_1 = \sqrt{(x_2 + x_1)^2 - 4x_2x_1}$$

$$= \sqrt{16 - 4xy}$$

$$= \sqrt{16 - 4y}$$

c. (i) $SV = 4\pi^2 Sy$

(ii) Consider the part with base OAB:

$$\begin{aligned} V &= \int_0^8 4\pi^2 \cdot dy \\ &= \int_0^8 8y dy \quad \text{since } x^2 = 2y \\ &= 4[y^2]_0^8 \\ &= 256 \text{ m}^3 \end{aligned}$$

Consider the part with semicircular base:

The eqn of the circle containing this is

$$\begin{aligned} x^2 + (y-8)^2 &= 16 \quad (\text{Centre is } (0,8), r=4) \\ \text{i.e. } x^2 &= 16 - (y-8)^2 \end{aligned}$$

$$\begin{aligned} \therefore V &= \int_8^{12} 4 [16 - (y-8)^2] dy \\ &= 4 \left[16y - \frac{1}{3}(y-8)^3 \right]_8^{12} \\ &= 4 \left[16 \times 12 - \frac{1}{3} \times 4^3 - 16 \times 8 \right] \\ &= 170\frac{2}{3} \text{ m}^3 \end{aligned}$$

$$\begin{aligned} \therefore \text{Volume of solid} &= 256 + 170\frac{2}{3} \\ &= 426\frac{2}{3} \end{aligned}$$

Q.3 a. $\int \frac{dx}{\sqrt{x^2+4x+3}} = \int \frac{dx}{\sqrt{(x+2)^2-1}}$

Let $u = x+2$

$$= \int \frac{du}{\sqrt{u^2-1}}$$

$$= \ln \{u + \sqrt{u^2-1}\} \quad (\text{table of standard int.'s})$$

$$= \ln \{(x+2) + \sqrt{x^2+4x+3}\}$$

b. $u = e^x$
 $du = e^x \cdot dx$
 $\text{or } dx = \frac{du}{u}$

$$I = \int \frac{1}{u + \frac{1}{u}} \cdot \frac{du}{u}$$

$$= \int \frac{du}{u^2+1}$$

$$= \tan^{-1} u + c$$

$$= \tan^{-1} e^x + c$$

c. $t = \tan \frac{x}{2}$
 $\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$
 $= \frac{1}{2} (1 + \tan^2 \frac{x}{2})$
 $= \frac{1}{2} (1 + t^2)$

$$dx = \frac{2 dt}{1+t^2}$$

$$x=0, t=0$$

$$x=\frac{\pi}{2}, t=1$$

$$I = \int_0^{\frac{\pi}{2}} \frac{1}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \times \frac{2dt}{1+t^2}$$

$$= \int_0^{\frac{\pi}{2}} \frac{2 dt}{1+t^2+1-t^2+2t}$$

$$= \int_0^{\frac{\pi}{2}} \frac{2 dt}{2+2t}$$

$$= \int_0^{\frac{\pi}{2}} \frac{dt}{1+t}$$

$$= [\ln(1+t)]_0^1$$

$$= \ln 2$$

d. Let $\frac{6x+4}{(x+2)(4-3x^2)} = \frac{A}{x+2} + \frac{Bx+C}{4-3x^2}$

$$6x+4 \equiv A(4-3x^2) + (Bx+C)(x+2)$$

Put $x=-2 \Rightarrow -8 = A(4-12) \Rightarrow A=1$

$x=0 \Rightarrow 4 = 4A+2C \Rightarrow C=0$

$x=1 \Rightarrow 10 = A+3B+3C \Rightarrow B=3$

$$I = \int \left(\frac{1}{x+2} + \frac{3x}{4-3x^2} \right) dx$$

$$= \ln(x+2) - \frac{1}{2} \ln(4-3x^2) + C$$

e. $\int \tan^4 x \, dx = \int \tan^2 x (\sec^2 x - 1) \, dx$
 $= \int \tan^2 x \sec^2 x \, dx - \int \tan^2 x \, dx$
 $= \frac{1}{3} \tan^3 x - \int (\sec^2 x - 1) \, dx$
 $= \frac{1}{3} \tan^3 x - \tan x + x + C$

f. (i) $I_n = \int_1^e (\ln x)^n \cdot \frac{d}{dx} \left(\frac{x^2}{2} \right) \cdot dx$
 $= \left[\frac{x^2}{2} (\ln x)^n \right]_1^e - \int_1^e \frac{x^2}{2} \times n (\ln x)^{n-1} \cdot \frac{1}{x} \, dx$
 $= \frac{e^2}{2} - \frac{n}{2} \int_1^e x (\ln x)^{n-1} \, dx$
 $= \frac{e^2}{2} - \frac{n}{2} I_{n-1}$

$$\begin{aligned}
 \text{f. (ii)} \quad I_3 &= \frac{e^2}{2} - \frac{n}{2} [I_2] \\
 &= \frac{e^2}{2} - \frac{n}{2} \left[\frac{e^2}{2} - \frac{n}{2} I_1 \right] \\
 &= \frac{e^2}{2} \left(1 - \frac{n}{2} \right) + \frac{n^2}{4} I_1 \\
 &= \frac{e^2}{4} (2-n) + \frac{n^2}{4} \left[\frac{e^2}{2} + I_0 \right] \\
 &= \frac{e^2}{4} \left(2-n + \frac{n^2}{2} \right) + I_0
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } I_0 &= \int_1^e x \cdot dx \\
 &= \left[\frac{x^2}{2} \right]_1^e \\
 &= \frac{e^2}{2} - 1
 \end{aligned}$$

$$\begin{aligned}
 \therefore I_3 &= \frac{e^2}{8} (4-2n+n^2) + \frac{e^2}{2} - 1 \\
 &= \frac{e^2}{8} (4-2n+n^2+4) - 1 \\
 &= \frac{e^2}{8} (n^2-2n+8) - 1
 \end{aligned}$$